

Using Machine Learning to Map Cell Lineage from Gene Expression Data

Abstract

Cancer has been a major societal, public health, and economic challenge in the 21st century, ranking as the second leading cause of death in the United States overall. In 2025 alone, an estimated 2.05 million new cancer cases and 620,000 deaths are projected to occur in the United States. With cancer being a prevalent health and economic issue, there has been a major effort in identifying genes contributing to the proliferation of different cancer types. However, the pronounced molecular heterogeneity of tumors complicates the identification of single “driver” genes, motivating systematic efforts to map lineage-specific gene programs. The Cancer Cell Line Encyclopedia (CCLE) is a data repository of gene expression across hundreds of cancer lineages. Our goal is to analyze the large bulk of available expression data and see if expression levels can be traced across cell lineage. We trained a machine learning model to predict cell type from expression levels. Then using measures of feature importance (e.g. SHAP values), we determined which genes have the largest effect on model prediction and might be enriched in cancer. Additionally, we found genes that have high differential expression, suggesting they are specifically enriched in only a single cancer lineage. This analysis uncovers the relative importance of genes implicated in cancer. Identification of lineage-defining genes in cancer could guide targeted therapies and enhance our understanding of tumor evolution. Furthermore, these insights support oncology by linking gene expression patterns to cell identity.

Introduction

Cancer remains the second leading cause of death in the United States, with approximately 2.05 million new cases and 620,000 deaths projected for 2025 alone ¹. Despite decades of research, cancer continues to present one of the greatest economic and biomedical challenges of the 21st century. The term “cancer” encompasses a wide array of subtypes affecting nearly every organ system, each characterized by unique genetic, epigenetic, and molecular profiles. This diversity contributes to the complexity of diagnosis, prognosis, and treatment, underscoring the need to isolate and identify the genes that drive each cancer subtype.

A major obstacle in cancer biology is its genetic and molecular diversity. Even tumors arising from the same tissue can differ substantially in their gene expression profiles. This complexity complicates the identification of single “driver” genes responsible for tumor initiation and progression. Consequently, researchers have shifted toward large-scale, systematic efforts to map lineage-specific gene expression profiles that could characterize distinct cancer types. Understanding gene expression has been fundamental to identifying potential genetic “drivers” of cancer. For example, the discovery of the *BRCA1* gene more than 30 years ago was a landmark in cancer genetics ²; it revealed a key genetic factor associated with hereditary breast and ovarian cancer and dramatically advanced women’s health by enabling early screening and risk reduction strategies.

In recent years, the major effort of large-scale multi-omics research across diverse cancer subtypes has revolutionized cancer biology and expanded our understanding of the genetic drivers underlying tumor development. The Cancer Cell Line Encyclopedia (CCLE) ³ provides comprehensive genetic profiling of over a thousand human cancer cell lines ^{4,5}. This resource includes transcriptomic, proteomic, and other sequencing data that enable researchers to systematically examine how gene expression varies across tissues and tumor types. By leveraging such data, it becomes possible to investigate expression patterns

that may help predict driver genes for different cancer lineages and to identify lineage-defining genes that could serve as biomarkers or therapeutic targets.

Simultaneously, advances in machine learning (ML) have provided powerful tools for analyzing high-dimensional biological data. ML methods can learn complex, nonlinear relationships between gene expression patterns and phenotypic traits, such as tumor lineage or cancer subtypes. When coupled with clustering and feature importance quantification methods, such as SHAP (SHapley Additive exPlanations) ⁶ values, these models can highlight potential specific genes that might strongly contribute to particular cancer subtype outcomes, revealing potential lineage-specific regulators or oncogenic pathways.

In this project, we utilize gene expression data from the CCLE to explore expression patterns that may isolate and identify potential gene drivers for cancer lineages. Specifically, we apply unsupervised and supervised learning approaches to visualize and cluster gene expression profiles across cancer cell lines. Dimensionality reduction techniques, such as t-distributed stochastic neighbor embedding (t-SNE), were used to identify clustering of cancer cell lineages in gene expression space, while hierarchical clustering was used to detect potential lineage-specific gene expression patterns that may reveal genes or groups of genes driving particular cancer types. Additionally, we performed a deep dive into gene-level patterns by analyzing the expression of the top two predicted most important genes that indicate lineage. To complement this analysis, we conducted a literature review to investigate the biological roles and molecular processes associated with these genes, providing context for their potential involvement in cancer development and progression. By linking computational patterns with biological interpretation, our goal is to uncover potential gene drivers that contribute to and define distinct cancer cell lineages. Ultimately, this integrative analysis aims to improve our understanding of the genetic contributions underlying lineage specificity and to utilize machine-learning frameworks to reveal relationships between gene expression and tumor identity.

Overall Goals and Aims

The Cancer Cell Line Encyclopedia (CCLE) ³ provides an extensive resource of gene expression data across hundreds of human cancer cell lines ^{4,5}. By leveraging this dataset, it becomes possible to explore how gene expression varies across cancer lineages and identify genes that are characteristic of specific tumor types. In this project, we used machine learning models trained on CCLE expression data to predict cancer lineage from gene expression patterns. Using interpretability metrics such as SHAP⁶ values, we identified genes with high predictive power and determined which are enriched within particular cancer lineages. Ultimately, this approach could help reveal lineage-specific genes and provide insights into the molecular underpinnings of cancer, guiding more targeted and effective therapeutic strategies.

Results & Discussion

Global Transcriptional variability suggests lineage-associated expression patterns

To evaluate the extent to which transcriptional variation captures cancer cell lineage identity, we first examined the 50 genes with the greatest variance across all cancer lineages in the DepMap expression dataset. The resulting heatmap (**Figure 1**) demonstrates potential lineage-associated expression patterns despite the limited gene subset. Several genes exhibit elevated expression in specific lineages; for example, SPARC, S100A6, VIM, FN1, and COL1A1 show strong expression in fibroblast/mesenchymal-type lineages, consistent with their roles in fibroblast extracellular-matrix biology and mesenchymal identity ⁷⁻⁹. However, high expression alone does not necessarily indicate that

these genes act as causal lineage drivers. Rather, substantial differences in gene-expression levels may reflect underlying lineage-specific biology or differentiation states. For instance, although S100A6 displays high expression across many cancer lineages and has been implicated in epithelial-mesenchymal transition and cell migration ¹⁰, its comparatively lower expression in eye, peripheral nervous system, and testis lineages may suggest tissue-restricted regulatory program. These observations align with previous studies demonstrating that subsets of high-variance genes tend to capture broad tissue-of-origin patterns in cancer cell lines ³.

Hierarchical clustering of the top 1,000 most variable genes reveals biologically coherent lineage groupings

To capture a broader and more representative transcriptional landscape, hierarchical clustering was performed on the top 1,000 most variable genes across cancer lineages (**Figure 2**). This analysis organizes cancer lineages based on their global gene-expression similarity. The dendrogram reveals clear grouping of biologically related cancer types. For example, lymphoid and myeloid lineages cluster closely together, consistent with their shared hematopoietic origin and the observation from large pan-cancer studies that hematologic malignancies form a distinct molecular class.¹¹ Neural lineages, including CNS/brain and peripheral nervous system cancers, also occupy neighboring branches, reflecting related neuroectodermal developmental programs and conserved neuronal gene-expression networks noted in transcriptomic atlases of nervous system tumors ¹².

Epithelial cancers form an extended grouping that includes breast, lung, ovary/fallopian tube, gastrointestinal, and bladder/urinary tract lineages. This pattern is consistent with pan-cancer analyses from TCGA showing that many epithelial tumors share transcriptomic programs related to cell-cell junctions, intermediate filaments, and cytokeratin biology, and often cluster together in molecular space despite originating from different organs ^{11,13}. In contrast, lineages such as testis and thyroid appear as outliers on longer dendrogram branches, likely reflecting highly specialized gene-expression programs (for example, germ-cell-specific signatures in testicular tumors) that differ substantially from those of most somatic epithelial and mesenchymal tissues ¹⁴.

Hierarchical clustering of the top 500 most variable genes reveals potential co-expressed genes

To identify co-expression patterns among genes rather than among cancer lineages, hierarchical clustering was performed on the top 500 most variable genes across all 26 lineages in the expression dataset (**Figure 3**). In this analysis, each gene was clustered according to its expression pattern across the 26 lineages, allowing visualization of gene expression that tends to vary together across tissues. The resulting dendrogram reveals several large and biologically coherent clusters of genes, suggesting the presence of structured regulatory co-expressions rather than random high-variance noise.

The modular structure seen in the hierarchical clustering of gene expression (**Figure 3**) mirrors findings from large-scale cancer transcriptome studies. Pan-cancer analyses from The Cancer Genome Atlas (TCGA) showed that groups of co-expressed genes form robust expression “signatures” that largely follow tissue-of-origin and major biological processes such as epithelial differentiation, immune infiltration, and stromal activation ¹³. Likewise, co-expression analyses in breast cancer identified gene modules whose coordinated activity captures clinically relevant heterogeneity and functional “hallmarks”

of tumors, reinforcing the idea that gene-level clustering reveals biologically meaningful networks rather than artifacts ¹⁵.

The hierarchical clustering of the top 500 most variable genes supports the view that transcriptional variability in this dataset is highly structured. High-variance genes tend to fall into coherent, biologically interpretable modules corresponding to epithelial identity, stromal/immune programs, and other functional pathways. Rather than representing noise, these modules likely capture lineage-linked processes such as differentiation state, cytoskeletal and junctional organization, and microenvironment-related signaling. As such, the co-expression patterns uncovered here provide a useful basis for downstream machine-learning analyses, for predicting lineage from gene expression and for prioritizing candidate lineage-associated or pathway-level genes for further study.

t-SNE visualization identifies clustering of cancer lineages

To evaluate lineage structure using a nonlinear dimensionality-reduction approach, t-distributed Stochastic Neighbor Embedding (t-SNE) was applied to the median log-transformed gene-expression profiles from all cancer lineages in the dataset (**Figure 4**). Unlike hierarchical clustering, which emphasizes global similarity across all samples, t-SNE preserves local neighborhood structure and is widely used to visualize transcriptional similarity in high-dimensional biological datasets, including single-cell RNA-seq and bulk tumor profiles. This method is especially effective for uncovering nonlinear manifolds and local clusters in complex genomic data.

The t-SNE embedding revealed distinct, lineage-specific clusters. Notably, breast, bowel, myeloid, and lymphoid lineages formed well-defined regions in the reduced 2D space, indicating consistent intra-lineage transcriptional similarity. Neural lineages, including CNS/brain and peripheral nervous system cancers, also clustered cohesively, supporting the patterns observed in **Figure 2**. In contrast, several epithelial lineages (including lung, head and neck, bladder/urinary tract, and esophagus/stomach) displayed partial overlap, reflecting their shared developmental pathways and molecular programs. Such overlap has been reported in multiple pan-cancer transcriptomic studies, where epithelial tumors cluster along gradients rather than forming discrete islands.¹¹ The overlap observed in t-SNE space here is consistent with those findings and likely reflects both shared developmental origins and convergent epithelial gene-expression programs.

Together, these observations demonstrate that lineage-defining transcriptional structure is preserved across both hierarchical and nonlinear dimensionality-reduction methods. The agreement between t-SNE clustering and lineage supports the validity of using gene-expression signatures for lineage prediction and suggests that there are underlying patterns for identifying lineage-associated genes that could be learned and harnessed by machine learning models.

Cancer lineage is predicted by gene expression

In order to capture the patterns observed by clustering and predict cancer cell lineage from gene expression, we trained a machine learning model. We first removed all lineages with less than 10 occurrences and split the remaining data into 5 folds, with fold 4 held out as a test set, and folds 0-3 used in 4-fold cross validation training. This ensured at least 2 samples of each lineage in each fold. We used a gradient-boosted decision forest implemented in LightGBM ¹⁶ as our classification model. For each

training set, we kept the gene expression of the top 10% most variant as the input features of the model. The average accuracy on the test fold was 74.8% across all four cross-validated models. Using an ensemble of the models, by averaging predicted class probabilities, further boosted test accuracy to 77.9% (**Figure 5**). Examining accuracy by cell lineage reveals that cell lineage performance negatively correlates with standard deviation with higher performing cell lineages showing less variance than lower performing ones (Pearson $r = -0.70$, **Figure 6**). This is likely due to the small number of some cell lineages. Accuracy is correlated with the number of samples in the lineage (Pearson $r = 0.58$, **Figure 6**). The high accuracy of the model on this task suggests that it sufficiently captures gene expression patterns of cancer cells and how these expression patterns differentiate between lineages.

Predictive models can capture genes linked to cancer

In order to further interpret the model and validate that it is capturing biologically relevant information, we did a SHAP ⁶ (SHapley Additive exPlanations) value analysis to calculate the most important genes for prediction. SHAP values offer a model-agnostic way to calculate feature importance by determining how much each feature contributed to the final prediction for a given sample. For each cross-validated model, we calculated the SHAP values of each gene using a subset of 1000 samples from each model's training set. We then averaged the absolute value of the SHAP values across all models as a measure of feature importance. Genes whose expression levels resulted in higher mean absolute SHAP values were deemed most important. The top 25 most important genes can be found in **Figure 7**.

COL9A2

Based on the feature importance models, COL9A2 emerged as the gene with the most predictive power for lineage classification. Although COL9A2 expression shows modest variation across cell lineages, there is substantial overlap in the overall distribution, with most lineages exhibiting minimum values near zero (**Figure 8**). This overlap makes it difficult to determine whether COL9A2 overexpression is truly lineage-specific or instead reflects a broader feature of tumor biology. Furthermore, the uneven sample sizes across cancer cohorts (**Figure 10**) complicate interpretation, raising questions about whether the expression of a single gene can adequately account for lineage differences. Importantly, the observation that many lineages display near-zero minimum expression suggests that COL9A2 activity may be restricted to particular tumor contexts or microenvironmental states, rather than serving as a universal marker.

The COL9A2 gene encodes alpha-2 chain of type IX collagen, one of the three subunits that form a heterotrimeric collagen complex ¹⁷. Type IX collagen is a critical structural component of cartilage, where it interacts with type II collagen fibrils to stabilize the extracellular matrix ¹⁸. Mutations in COL9A2 are associated with various connective tissue disorders including Stickler syndrome, epiphyseal dysplasia, and disc degeneration ^{17,19,20}. These conditions highlight the significance of COL9A2 in maintaining cartilage integrity and skeletal development.

In the context of cancer, data from The Protein Atlas indicate that COL9A2 is classified as prognostic for renal cancer types ²¹. Additionally, collagen remodeling is increasingly recognized as a driver of tumor progression, influencing processes of cell adhesion, migration, invasion, and angiogenesis ²². Overexpression of collagen genes has been reported in several cancers, including gastric cancer, where altered collagen deposition and signaling contribute to tumor growth and metastasis ²³. While COL9A2

itself has not been as extensively studied as collagens like COL1A1 or COL4A1, its structural role in the extracellular matrix suggests it may participate in similar pathways of tumor microenvironment. Importantly, our predictive models highlight COL9A2 as a gene with strong lineage-specific signal, underscoring their ability to detect potentially relevant molecular features which aligns with current clinical resources, such as the Protein Atlas. This demonstrates the value of these models in uncovering candidate genes that may contribute to lineage-specific tumor biology and in guiding future functional investigations.

SNAPC1

After investigating COL9A2, we also decided to look at the gene with the second highest importance for our predictive model: small nuclear RNA activating complex 1, or SNAPC1. We followed the same protocol as COL9A2 and found that the gene seems to have many overlaps in expression when looking across different cancer lineages (**Figure 9**). While this was the second most important gene for our machine learning model in predicting cell lineage, it is not within the top 50 genes with the most variance in our dataset (**Figure 13**). This implies that our model is looking at underlying trends within the data, and potentially expression interactions between genes, in order to make its predictions rather than simply looking at variance.

That said, we hoped to see if there was any significance in SNAPC1 expression in cancers in general. We found that there already was some literature available that characterized its involvement as a tumor formation associated gene, and plays a role as a general transcriptional coactivator in the phosphorylation of S2 on RNA Polymerase during the elongation phase of translation²⁴. Interestingly, The Protein Atlas²⁵ also mentions that SNAPC1 has low cancer specificity, meaning that the gene is not significantly up or down regulated in different types of cancers, which reflects our own observations from Figure 9. Regardless, the presence of this gene in the top SHAP features suggests that the model is capturing meaningful patterns in the data, and that it is able to find genes that are relevant to cancer proliferation for making predictions.

FXVD3

We performed a t-SNE on the top 50 SHAP genes and observed some clustering, indicating that just using expression from the top 50 genes was sufficient clustering some cancer lineages (**Figure 12**). After looking at the top two SHAP genes and performing the t-SNE, we were curious to see if there was any overlap between the top 50 genes with the most variance and the top 50 SHAP genes, but we only found one gene, FXVD3, that was shared between the sets. We looked at the expression of this gene across lineages and did observe a higher level of differential expression compared to COL9A2 and SNAPC1 (**Figure 11**). This leads us to believe that the model is able to extrapolate information from expression data beyond simply using variance to make predictions.

Conclusion

Leveraging the large-scale gene expression data from the Cancer Cell Line Encyclopedia, we show that cancer cell lineages transcriptional clusters exhibit a potential transcriptional signature that could highlight potential gene drivers for specific cancer lineages. Mapping high-variance genes and hierarchical clustering revealed biologically relevant groupings of lineages and gene modules that could

suggest co-expression, while nonlinear dimensionality reduction demonstrated some lineage-specific organization of expression space. These patterns enabled lineage prediction using a gradient-boosted machine-learning model, which achieved an ensemble accuracy of 77.9%, indicating that lineage identity can be engrained in gene-expression profiles.

Model interpretability analyses revealed that the features driving prediction extend beyond simple variance. The ML model was able to leverage differential expression genes related to transcriptional regulation (like SNAPC1) to distinguish cancer (as in this gene had a really high SHAP value). SHAP-based feature importance identified genes such as COL9A2, SNAPC1, and FXYP3 as influential, including genes that do not appear among the most variable, suggesting the model captures deeper, non-linear gene expression relationships. Together, these results demonstrate that ML approaches can predict global and subtle gene expression patterns to infer lineage and highlight potential lineage-associated gene drivers.

Overall, this approach provides a framework for identifying transcriptional programs that can be linked to cancer lineage identity. By combining hierarchical clustering, nonlinear-embedding t-SNE, and ML, we can identify expression signatures and candidate genes that may contribute to lineage specificity in cancer. These initial results highlight the power of computational approaches for uncovering nonlinear biological structures in high-dimensional omics data and underscores how such methods can advance the discovery of lineage specific and potential gene drivers that could be ideal candidates for therapeutics.

Figure Legends

Figure 1. *Heatmap of the top 50 most variable genes across cancer lineages.* Each row represents a cancer cell-line lineage, and each column corresponds to one of the top 50 genes with the highest variance in expression across all lineages. The heatmap displays the median log-transformed gene expression (TPM) for each lineage-gene pair. Expression levels are scaled from low (blue) to high (red), as indicated by the color bar (0–14).

Figure 2. *Hierarchical clustering of cancer lineages based on the top 1,000 most variable genes.* Each row represents a cancer cell-line lineage, and each column corresponds to one of the 1,000 genes with the highest variance in expression across all lineages. The heatmap displays the median log-transformed gene expression (TPM) for each lineage-gene pair, with expression levels scaled from low (blue) to high (green) as indicated by the color bar (0–14). The accompanying dendrogram depicts hierarchical clustering of lineages based on similarity in their global expression profiles, revealing distinct groupings that reflect shared tissue origins, such as clustering of hematopoietic (myeloid and lymphoid), epithelial, and neural lineages.

Figure 3. *Hierarchical clustering of gene expression based on the top 500 most variable genes.* Each row corresponds to one of the 1,000 genes with the highest variance in expression across all lineages, and each column represents a cancer cell-line lineage. The heatmap displays the median log-transformed gene expression (TPM) for each lineage-gene pair, with expression levels scaled from low (blue) to high (green) as indicated by the color bar (0–14). The accompanying dendrogram depicts hierarchical clustering of genes based on similarity in their global expression profiles across cancer cell lineages, revealing distinct groupings that reflect potential co-expression between genes that might give rise to certain cancer cell lineages.

Figure 4. *t-SNE visualization of lineage-specific gene expression clustering across cancer cell lines.* Each point represents a lineage-gene pair from the DepMap dataset, embedded using t-distributed stochastic neighbor embedding (t-SNE) applied to median log-transformed TPM expression values. Points are colored according to their Oncotree lineage classification. Distinct clusters corresponding to major cancer types, including breast, bowel, and myeloid lineages, are observed, indicating that gene expression patterns are organized by tissue of origin. These lineage-specific transcriptional clusters suggest that certain genes may play pivotal roles in the establishment or maintenance of specific cancer lineages.

Figure 5. *Test fold accuracy for each cross validated model.* The fold labeled in the x-axis corresponds to the model validated on the given fold, for fold 0 through fold 3 respectively: 77.0%, 72.4%, 73.8%, 75.9%. The orange dotted line is the accuracy of taking the average predicted probability for each class as an ensemble of all models (77.9%).

Figure 6. *Test fold accuracy by cell lineage.* A boxplot showing the distribution of accuracies on the test fold separated by cell lineage across all models. The orange dotted line represents the average performance of the ensemble model. The number of samples from each lineage is shown in parentheses on the y-axis.

Figure 7. *Top 25 most important genes.* The mean absolute value of SHAP values for each gene across all models. SHAP values were calculated from a subset of 1000 samples from each model's training data.

Figure 8: *COL9A2 gene expression distribution by cancer lineage.* Each box corresponds to the expression of COL9A2 on a specific cancer cell line, sorted by descending median value.

Figure 9. *Expression of gene SNAPC1 across cancer cell lines.* Each box corresponds to the expression of SNAPC1 on a specific cancer cell line, sorted by descending median value.

Figure 10. *Total observation counts by lineage.*

Figure 11. *FXYP3 Expression by Cancer Lineage.* Each box corresponds to the expression of FXYP3 on a specific cancer cell line, sorted by descending median value.

Figure 12. *t-SNE of top 50 SHAP values.* Same as Figure 4, but only looking at the top 50 genes with the highest SHAP values.

Figure 13. *Venn Diagram of 50 highest SHAP and 50 highest variance genes.*

Figures

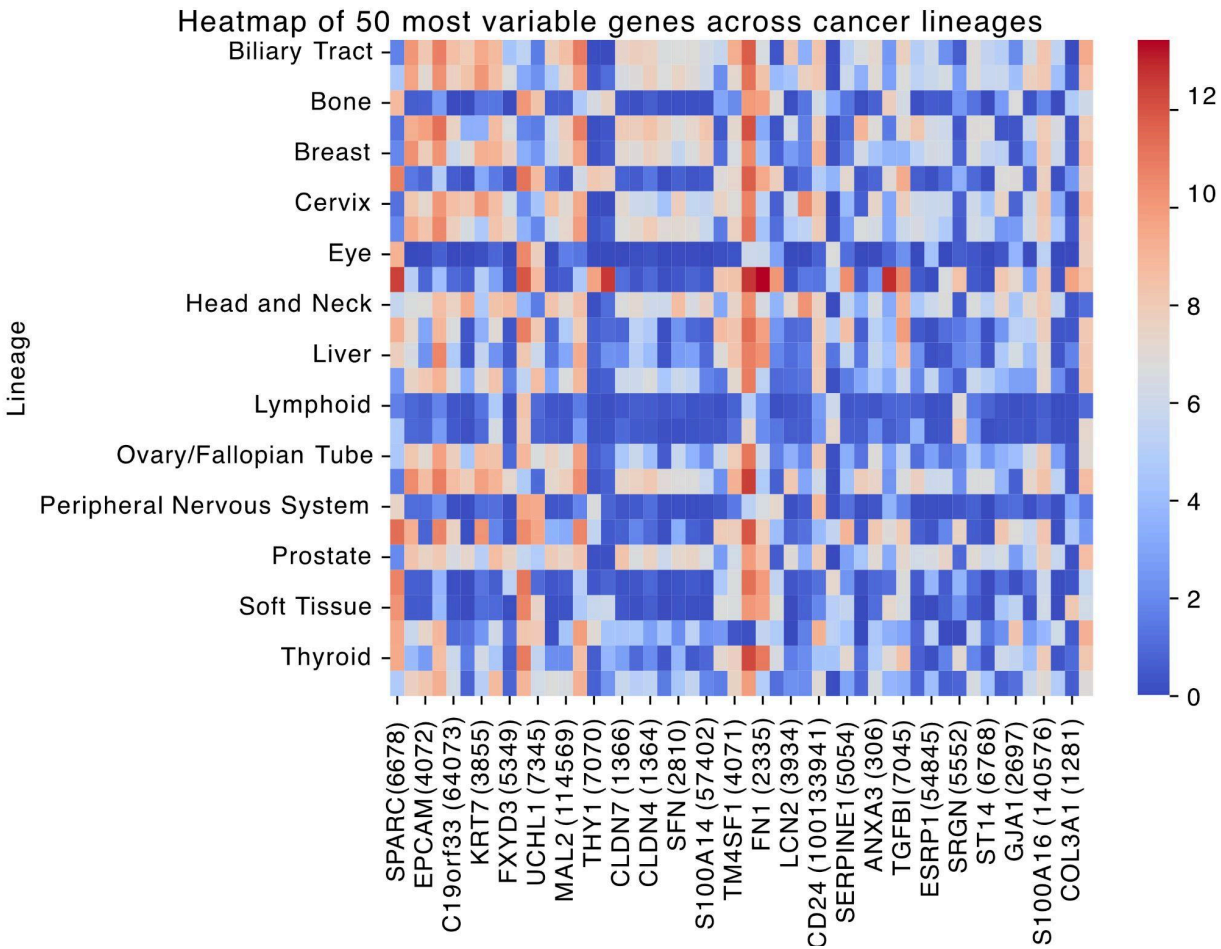


Figure 1. Heatmap of the top 50 most variable genes across cancer lineages.

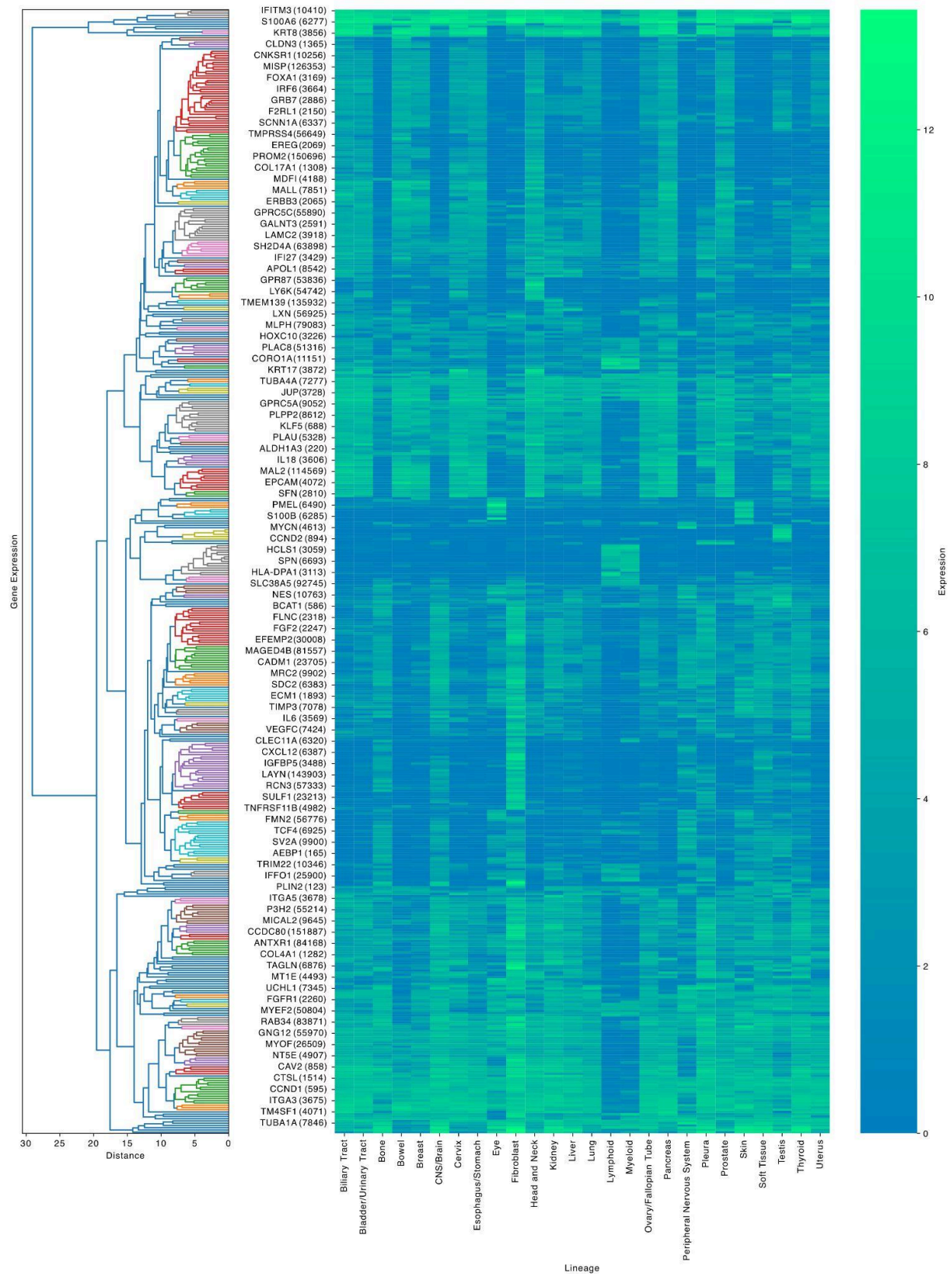


Figure 3. Hierarchical clustering of gene expression based on the top 500 most variable genes.

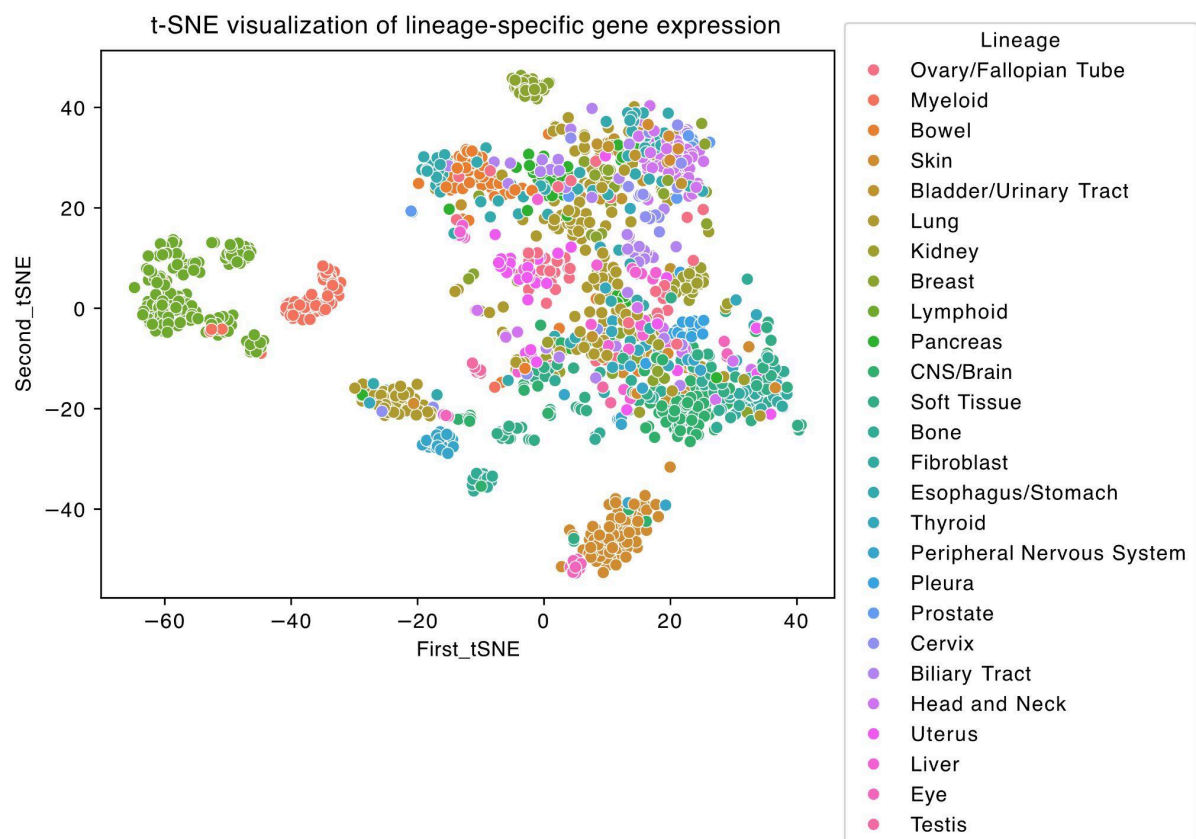


Figure 4. *t-SNE visualization of lineage-specific gene expression clustering across cancer cell lines.*

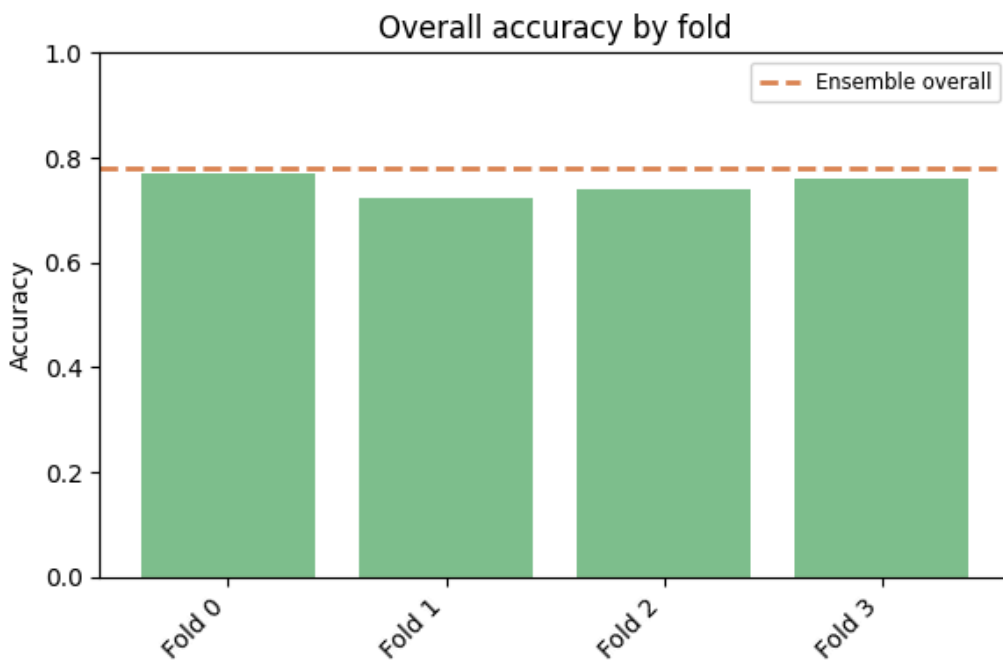


Figure 5. *Test fold accuracy for each cross validated model.*

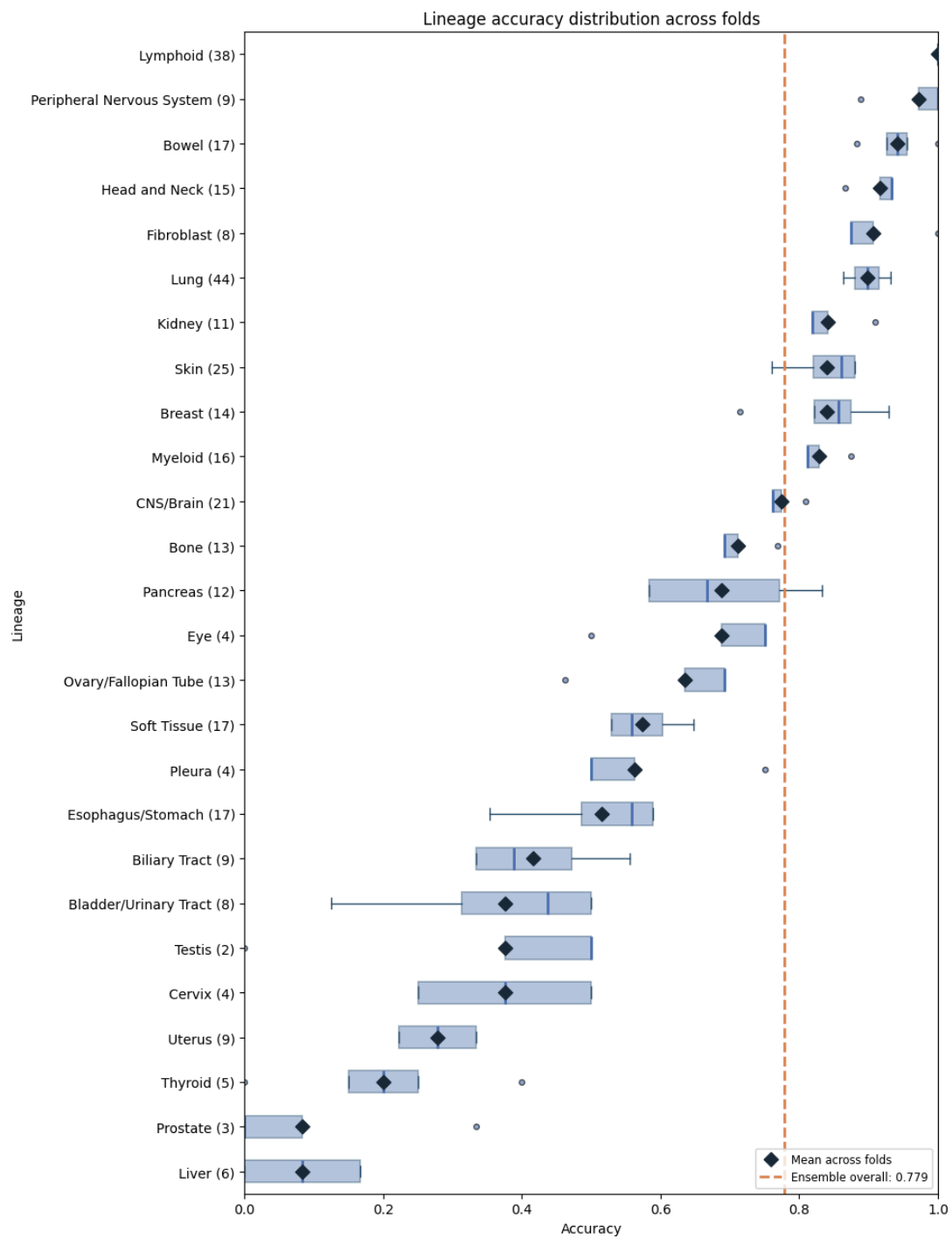


Figure 6. Test fold accuracy by cell lineage.

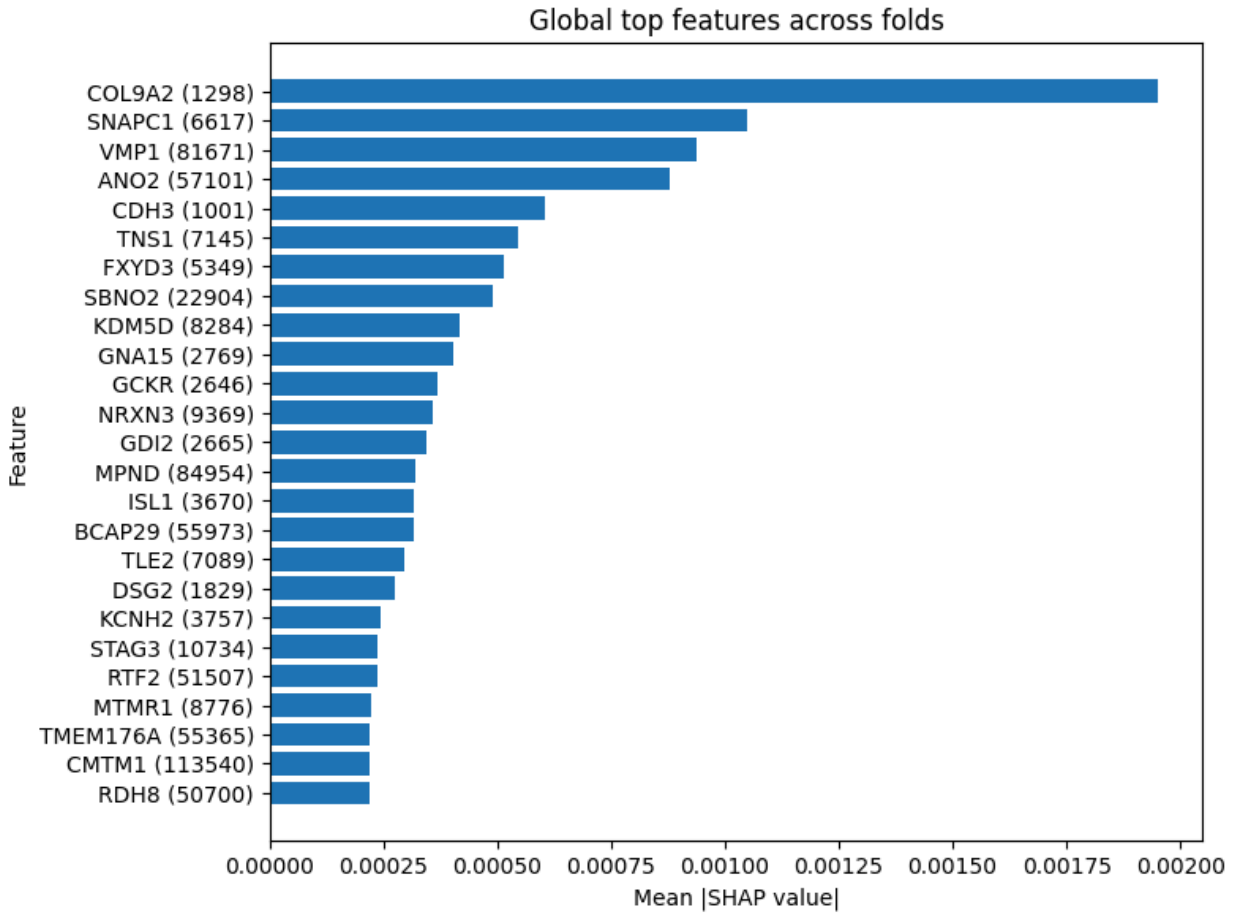


Figure 7. Top 25 most important genes.

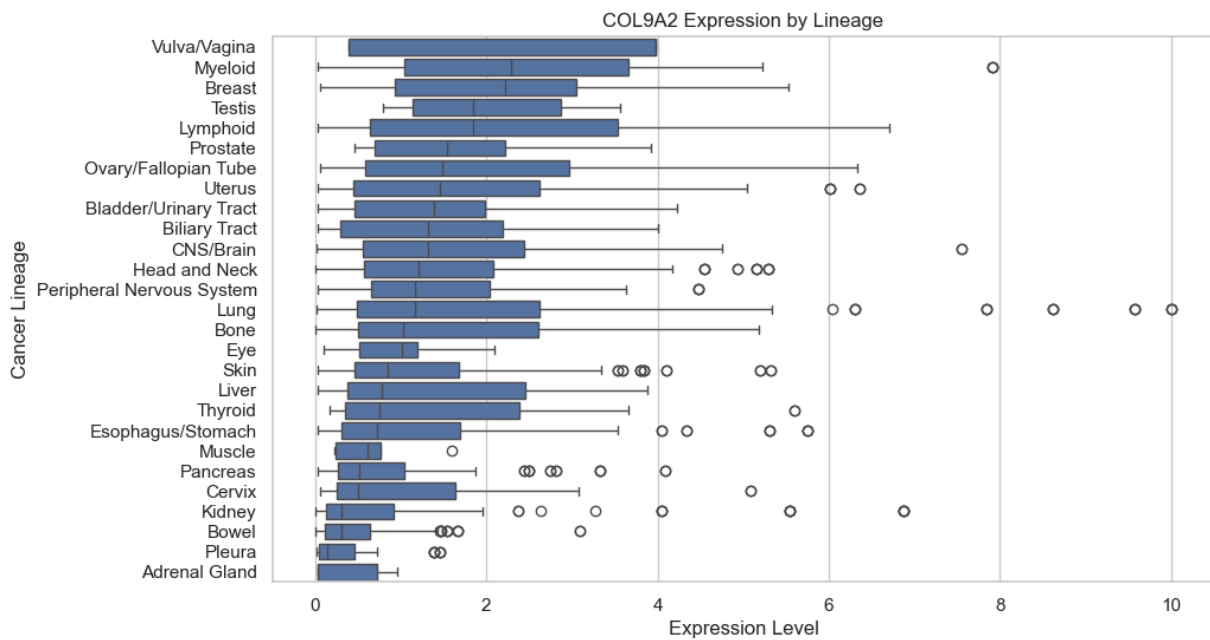


Figure 8. COL9A2 gene expression distribution by cancer lineage.

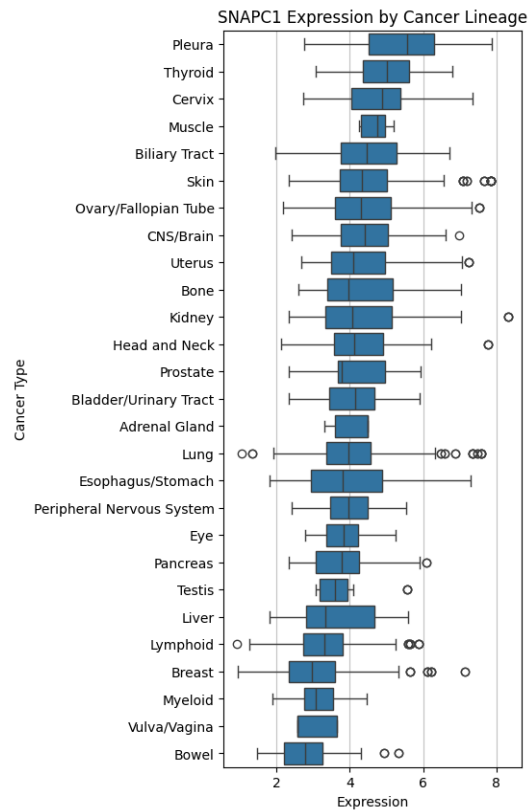


Figure 9. Expression of gene *SNAPC1* across cancer cell lines

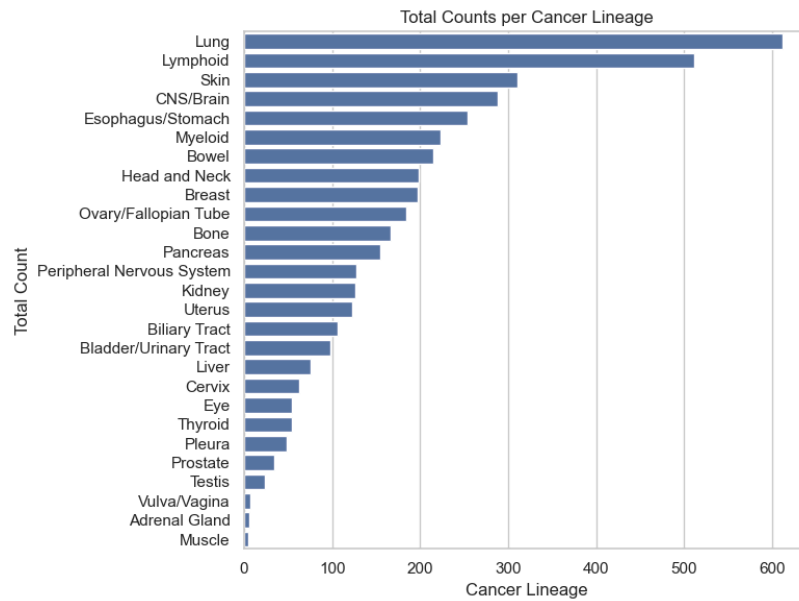


Figure 10. Total observation counts across cancer lineages.

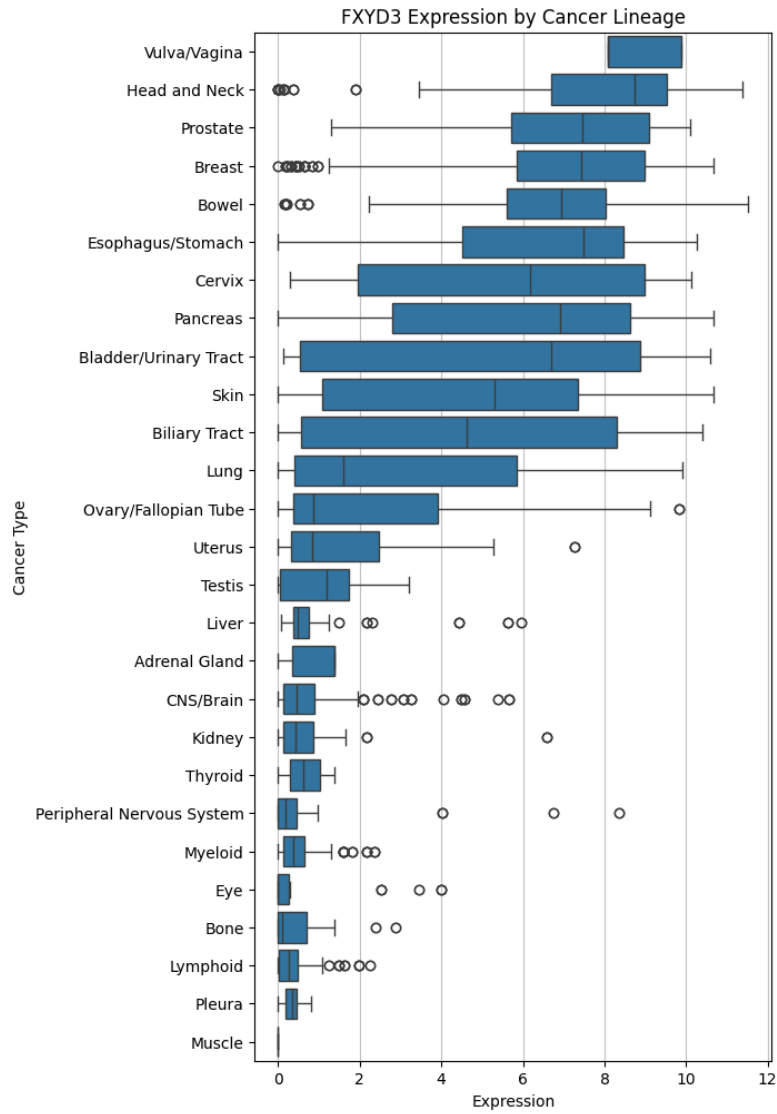


Figure 11. FXD3 Expression by Cancer Lineage

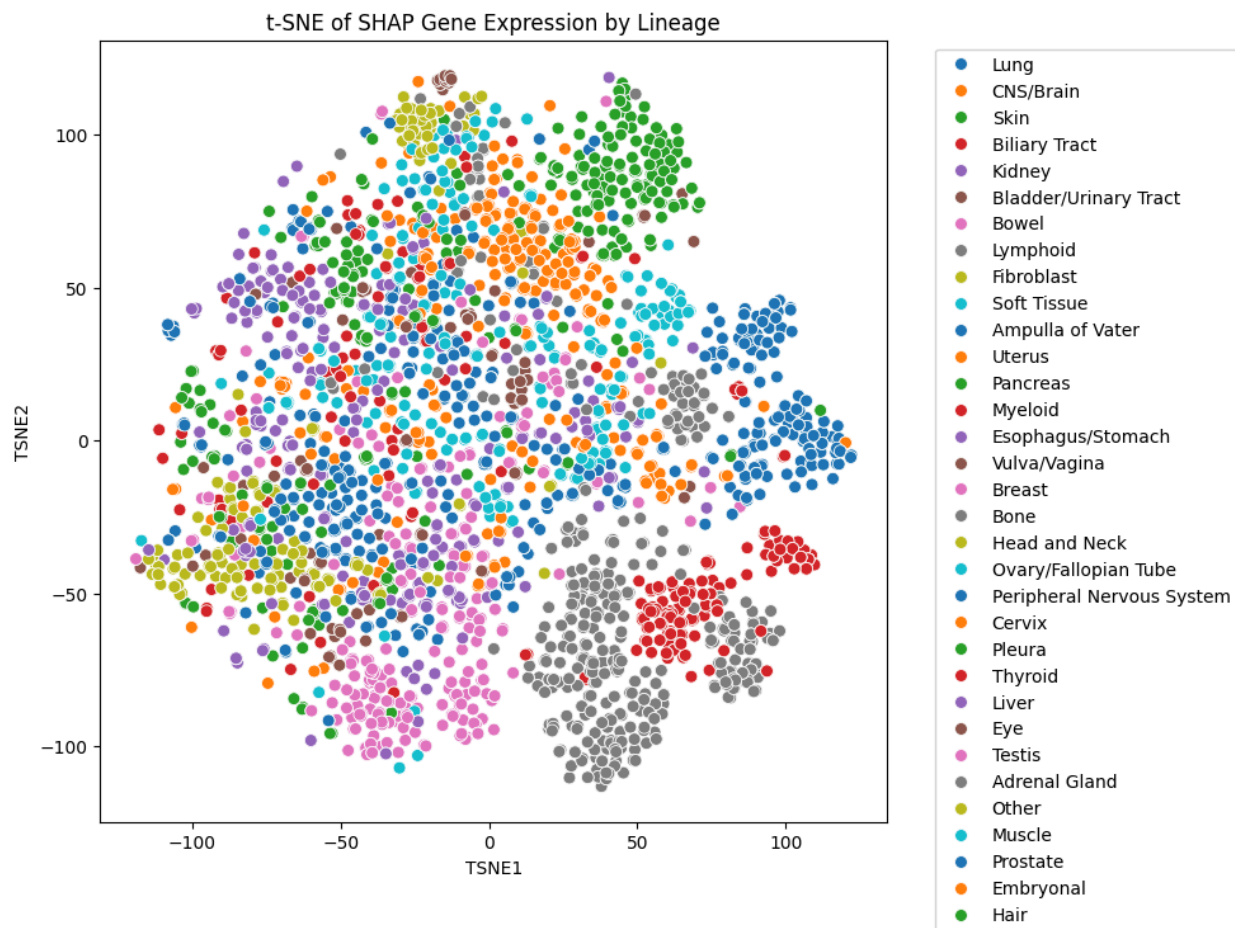


Figure 12. t-SNE of top 50 SHAP values.

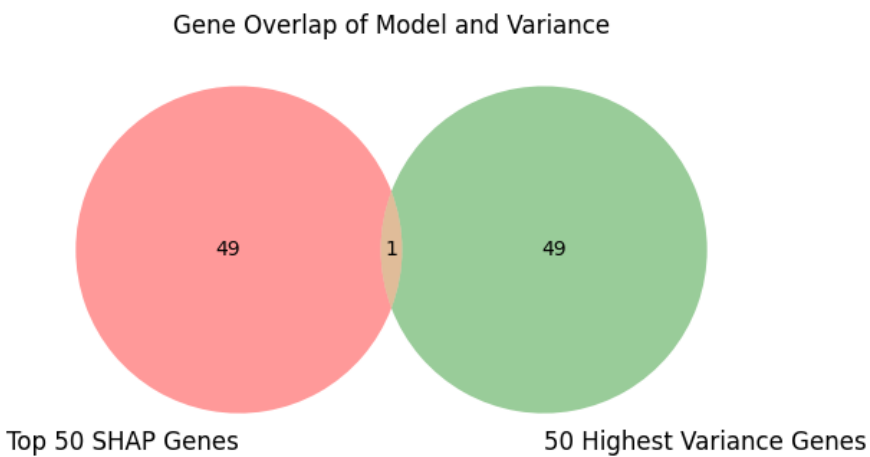


Figure 13. Venn Diagram of 50 highest SHAP and 50 highest variance genes.

Data Sources and Computational Methods

Datasets acquired from DepMap

All data was acquired from the Cancer Cell Line Encyclopedia and DepMap here: https://depmap.org/portal/data_page/?tab=allData. The relevant datasets include Model.csv which contains sample metadata, OmicsExpressionTPMLogp1HumanProteinCodingGenes.csv which contains gene expression data for all human genes indexed by a sequencing ID, and OmicsProfiles.csv which maps the sequencing ID to a cancer cell lineage.

Code

All code can be found on GitHub here: https://github.com/lpersyn/biostats_final_project/tree/main. Instructions for creating the environment and running the machine learning code are found in the README file (Figures 5-7). Notebooks for creating all other figures were run on Edupod.

Author Contributions

Gabriel created the heatmap, hierarchical clustering, and t-SNE (Figures 1-4). Logan coded, trained, and evaluated ML models (Figures 5 and 6) and also did SHAP analysis with help of AI (GitHub Copilot, Figure 7). Kristi looked at COL9A2 gene expression and total observations per lineage (Figures 8 and 10). Morgan looked at SNAPC1 and FXVD3, made the t-SNE for SHAP genes expression, and the venn diagram (Figures 9 and 11-13). All members (Logan, Morgan, Kristi, and Gabriel) contributed to the final report.

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